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Project Title	High-Performance Market Basket Analytics with Compressed FP-Trees from Scratch
Week	Week 4

WEEK- 4 Technical Report

Task Name

High-Performance Market Basket Analytics with Compressed FP-Trees from Scratch

Abstract

Market Basket Analysis is a data mining technique used to discover relationships between products purchased together by customers. This project implements a High-Performance Market Basket Analytics system using Compressed Frequent Pattern Trees (FP-Trees) from scratch. FP-Trees efficiently store transaction data and reduce database scanning, enabling faster association rule mining. The system identifies frequent itemsets and generates valuable insights for business decision-making.

Introduction

Market Basket Analysis helps businesses understand customer purchasing behavior by finding products that are frequently bought together. Traditional algorithms such as Apriori require multiple database scans and generate many candidate itemsets, resulting in high computational costs. FP-Growth overcomes these limitations by using a compressed FP-Tree structure, which stores transaction information efficiently and discovers frequent patterns without candidate generation.

Objectives

- Analyze customer transaction data.
- Construct a compressed FP-Tree from transaction records.
- Identify frequent itemsets efficiently.
- Generate association rules between products.
- Improve performance compared to traditional methods.

Theory

FP-Tree (Frequent Pattern Tree)

An FP-Tree is a compact data structure that stores transaction data while preserving item association information.

FP-Growth Algorithm

The FP-Growth algorithm works in two steps:

1. Build the FP-Tree from transaction data.
2. Mine frequent patterns from the FP-Tree recursively.

Advantages

- Reduces memory usage.
- Requires only two database scans.
- Eliminates candidate generation.
- Faster than Apriori for large datasets.

Methodology

1. Collect transaction data.
2. Calculate item frequencies.
3. Remove infrequent items.
4. Sort items by frequency.
5. Build the FP-Tree.
6. Mine frequent itemsets.
7. Generate association rules.
8. Analyze results.

Python Code

```
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import fpgrowth
import pandas as pd
transactions = [
    ['Milk', 'Bread', 'Butter'],
    ['Milk', 'Bread'],
    ['Bread', 'Butter'],
    ['Milk', 'Butter'],
    ['Milk', 'Bread', 'Butter', 'Eggs']
```

```

]
te = TransactionEncoder()
te_array = te.fit(transactions).transform(transactions)

df = pd.DataFrame(te_array, columns=te.columns_)

frequent_itemsets = fpgrowth(df, min_support=0.4, use_colnames=True)
print("Frequent Itemsets:")
print(frequent_itemsets)

```

Dataset

Sample Transaction Dataset:

Transaction ID	Items
T1	Milk, Bread, Butter
T2	Milk, Bread
T3	Bread, Butter
T4	Milk, Butter
T5	Milk, Bread, Butter, Eggs

Results

- Milk and Bread are frequently purchased together.
- Bread and Butter show strong association.
- Milk, Bread, and Butter form a common purchase pattern.
- FP-Growth efficiently discovers frequent itemsets.

Applications

- Retail recommendation systems.
- Product placement optimization.
- Cross-selling strategies.
- E-commerce analytics.
- Customer behavior analysis.

Advantages

- High processing speed.
- Reduced memory consumption.
- Suitable for large transaction databases.
- Accurate association rule discovery.

Conclusion

The High-Performance Market Basket Analytics system using Compressed FP-Trees effectively discovers frequent purchasing patterns from transaction data. By utilizing the FP-Growth algorithm, the system achieves better performance than traditional association rule mining methods, making it suitable for modern retail and e-commerce applications.